

1. $A = x^2 - 3x - 5$, $B = x^2 - 3x + 1$, $C = x^2 - 4x - 3$ であるとき, 次の計算をせよ。

(1) $B + C$

答.

(2) $2A + 2C$

答.

(3) $-A + B + C$

答.

(4) $-A - B + C$

答.

(5) $2A - 3B + 2C$

答.

(6) $2A + 3B - 3C$

答.

(7) $3A - B - 3C$

答.

(8) $2A - 3B - C$

答.

2. 次の 1 次方程式を解け。

(1) $3x + 1 = 5x + 2$

答.

(2) $2x + 2 = -x + 6$

答.

(3) $2x + 2(-8x - 2) = 1$

答.

(4) $5 + 2(7x + 7) = x$

答.

(5) $9x + 3(-9x + 8) = -7$

答.

(6) $4x - 1 = -3x - \frac{1}{4}$

答.

(7) $\frac{7}{3}x - 5 = x - \frac{4}{3}$

答.

(8) $-2x - 2 = -x + 1$

答.

1. $A = x^2 - 3x - 5$, $B = x^2 - 3x + 1$, $C = x^2 - 4x - 3$ であるとき, 次の計算をせよ。

(1) $B + C$

答. $2x^2 - 7x - 2$

(2) $2A + 2C$

答. $4x^2 - 14x - 16$

(3) $-A + B + C$

答. $x^2 - 4x + 3$

(4) $-A - B + C$

答. $-x^2 + 2x + 1$

(5) $2A - 3B + 2C$

答. $x^2 - 5x - 19$

(6) $2A + 3B - 3C$

答. $2x^2 - 3x + 2$

(7) $3A - B - 3C$

答. $-x^2 + 6x - 7$

(8) $2A - 3B - C$

答. $-2x^2 + 7x - 10$

2. 次の 1 次方程式を解け。

(1) $3x + 1 = 5x + 2$

$-2x = 1$

答. $x = -\frac{1}{2}$

(2) $2x + 2 = -x + 6$

$3x = 4$

答. $x = \frac{4}{3}$

(3) $2x + 2(-8x - 2) = 1$

$2x - 16x - 4 = 1$

$-14x = 5$

答. $x = -\frac{5}{14}$

(4) $5 + 2(7x + 7) = x$

$5 + 14x + 14 = x$

$13x = -19$

答. $x = -\frac{19}{13}$

(5) $9x + 3(-9x + 8) = -7$

$9x - 27x + 24 = -7$

$-18x = -31$

答. $x = \frac{31}{18}$

(6) $4x - 1 = -3x - \frac{1}{4}$

両辺に 4 をかけて, $16x - 4 = -12x - 1$

答. $x = \frac{3}{28}$

(7) $\frac{7}{3}x - 5 = x - \frac{4}{3}$

両辺に 3 をかけて, $7x - 15 = 3x - 4$

答. $x = \frac{11}{4}$

(8) $-2x - 2 = -x + 1$

$-x = 3$

答. $x = -3$